University of Colombo School of Computing

Application Laboratory

FreePlane Mind Mapping Tutorial

Learning Mind Mapping through a Detective Case Study

Session Duration: 60 minutes

Part 1: Step-by-step FreePlane practice (15–20 minutes)
Part 2: Guided mind-mapping activity – *The Case of the Missing Painting* (40–45 minutes)

Target Audience: Undergraduate Students

Tool: FreePlane (Open Source Mind Mapping Software)

Session Overview

This is a 1-hour hands-on laboratory session designed to introduce students to **FreePlane**, a free and open-source mind mapping tool. The session is split into two parts:

1. **Guided Practice (15–20 min):** Students individually work through a step-by-step practice sheet on their own machine, following exact instructions to learn FreePlane's core features.
2. **Applied Activity (40 min):** Students apply what they practiced to solve *The Case of the Missing Painting* – a mystery scenario that requires building a full mind map with branches, cross-links, icons, colors, and notes.

Learning Outcomes: By the end of the session, students should be able to create nodes and hierarchies, format nodes with color/icons/style, attach notes, create cross-links between non-adjacent nodes, and use a mind map as a reasoning and organization tool.

Software Version: These instructions are written for **FreePlane 1.x** (verified against the current stable release, 1.13.x) and use only menu items and shortcuts that have been stable across the 1.x release line.

1 Part 1: Step-by-Step Practice (15–20 minutes)

1.1 Instructions to Students

Work through this on your own, step by step.
Follow each numbered instruction exactly and check the result before moving to the next one. This lab computer has **no internet access**, so everything you need is written out below – there is no need to search online for help. You have **15–20 minutes**.

1.2 Step-by-Step Practice Sheet

Step 1 – Create a new map.

1. Open FreePlane from the desktop/start menu.
2. Go to `File` → `New`. A new map opens with a single central node called "New Mindmap."

Step 2 – Rename the central node.

1. Double-click the central node.
2. Delete the existing text and type: `My First Map`.
3. Press `Enter` (on the number pad or main keyboard) to confirm the text.

Step 3 – Add child nodes.

1. Click once on the central node to select it (single click, not double).
2. Press the `Insert` key. A new child node appears, ready for text.
3. Type `Idea 1` and press `Enter` to confirm.
4. Click the central node again, press `Insert`, and create `Idea 2`.
5. Click the central node again, press `Insert`, and create `Idea 3`.
6. You should now have three child nodes attached to the center.

Step 4 – Add a sibling node.

1. Click on `Idea 1` to select it.
2. Press `Enter`. This creates a new node at the *same level* as `Idea 1` (a sibling), not a child.
3. Type `Idea 1b` and press `Enter` to confirm.

Step 5 – Add a grandchild node.

1. Click on `Idea 2` to select it.
2. Press `Insert` to add a child under it.
3. Type `Sub-point A` and press `Enter`.

Step 6 – Move a node by dragging.

1. Click and hold on `Idea 1b`.
2. Drag it on top of `Idea 3` and release the mouse button.

3. `Idea 1b` should now be a child of `Idea 3` instead of a sibling of `Idea 1`.

Step 7 – Change node and branch color.

1. Right-click on `Idea 2`.
2. In the menu, go to `Format` → `Node Color` (or use the paint-bucket / "A" color icons on the toolbar if visible) and choose a color for the text.
3. Right-click `Idea 2` again and select `Format` → `Edge Color` to change the color of the branch line leading to it.

Step 8 – Add an icon.

1. Right-click on `Idea 3`.
2. Go to `Icons` in the right-click menu (or the `Icons` menu in the top menu bar).
3. Choose any icon, for example a star or flag. It will appear next to the node text.

Step 9 – Add a note to a node.

1. Click on `Sub-point A` to select it.
2. Go to `View` → `Note Panel` (or right-click the node and choose `Notes`) to open the note editing area, usually at the bottom or side of the screen.
3. Click inside the note area and type: `This is extra detail that doesn't need to be in the node itself.`
4. Click back on the node – notice a small note icon now appears next to it.

Step 10 – Create a cross-link between two nodes.

1. Click on `Idea 1` to select it.
2. Hold `Ctrl` and click on `Idea 3` as well, so both nodes are selected at the same time (`Idea 3` should be the *last* one you click – it becomes the arrow's target).
3. Right-click on either selected node and choose **Connect** from the context menu. The keyboard shortcut `Ctrl+L` does the same thing if you prefer not to right-click.
4. If your version doesn't show **Connect** in the right-click menu, look for it under the `Insert` menu in the top menu bar instead.
5. Another way that works on every version: hold `Ctrl+Shift` and drag `Idea 1` directly onto `Idea 3` with the mouse.
6. A curved arrow should now connect `Idea 1` and `Idea 3`, even though they are not directly connected in the tree structure. This is called a **cross-link** (or "connector").

Step 11 – Collapse and expand a branch.

1. Click the small circle at the edge of `Idea 2` (next to its child `Sub-point A`).
2. The branch collapses (hides its children). Click the circle again to expand it back.

Step 12 – Save your map.

1. Go to `File` → `Save As`.
2. Save the file as `firstmap.mm` in your student folder.

By the end of these 12 steps you will have practiced every feature needed for Part 2: creating nodes, siblings, and children; moving nodes; coloring nodes and branches; adding icons; adding notes; creating cross-links; collapsing branches; and saving a file.

2 Part 2: Applied Activity – The Case of the Missing Painting

2.1 Scenario Briefing

Setting: Overlook University's Fine Arts Gallery, during last night's Alumni Gala.
The Crime: *"Midnight Harbor,"* a rare 19th-century oil painting on loan from a private collector, vanished from the East Wing sometime between 9:00 PM and midnight. The frame was found empty, propped behind a curtain. No alarms were triggered – meaning whoever took it had access to the gallery's disable codes or a key.
Your Task: Build a FreePlane mind map that organizes the suspects, clues, locations, and timeline of the case. Use cross-links to show how individual clues connect to one or more suspects. At the end of the session, you must be able to name the thief and justify your answer using only your mind map.

2.2 The Four Suspects

1. **Professor Elena Vasquez** – Art History faculty, curated the exhibit. Has master key access. Was seen arguing with the gallery director earlier that week about the painting's insurance value.
2. **Marcus Chen** – Wealthy alumnus and known private collector. Attended the gala. Has bid on *"Midnight Harbor"* twice at past auctions and lost both times.
3. **Priya Patel** – Gallery security guard on duty that night. Knows the alarm disable codes. Claims she took a 20-minute break at 9:45 PM.
4. **Tom O'Brien** – Catering staff supervisor. Had full access to service corridors connecting to the East Wing. New hire, started 3 weeks ago.

2.3 The Clues

1. A muddy shoe print (size 10) found near the service corridor door.
2. The gallery's alarm log shows a 20-minute system disable at 9:45 PM.
3. A torn piece of black fabric caught on the East Wing window latch.
4. Professor Vasquez was seen on the main floor giving a speech from 9:30–10:00 PM.
5. Marcus Chen left the gala early, around 9:40 PM, telling friends he "had a headache."
6. Security footage from the front entrance was mysteriously looped/glitching from 9:40–10:05 PM.
7. Tom O'Brien was wearing a black catering uniform jacket that night.
8. Priya Patel's break-room swipe card log shows she badged *in* at 9:45 PM, not out.
9. A partial email draft found later, from an anonymous account, offering *"Midnight Harbor"* for private sale, timestamped 11:50 PM.
10. Marcus Chen's car was seen by a valet leaving the campus lot at 10:15 PM – later than his stated departure time.

2.4 Locations and Timeline

Locations	Main Hall (gala), East Wing (crime scene), Service Corridor, Break Room, Front Entrance, Parking Lot
Timeline	9:00 PM (gala starts) → 9:30 PM → 9:40 PM → 9:45 PM → 10:00 PM → 10:15 PM → 11:50 PM → Midnight

Note to students: Some clues are deliberately contradictory or partial red herrings (for example, Clue 4 seems to clear Professor Vasquez, while Clue 8 seems to incriminate Priya Patel but could simply mean her card malfunctioned). This is intentional – you must weigh conflicting evidence rather than build a single flat list.

2.5 Mind Map Requirements

Your finished FreePlane map must include **all** of the following:

1. A central node: **"The Case of the Missing Painting."**
2. Main branches for **Suspects**, **Clues**, **Locations**, and **Timeline**.
3. Each suspect as a child node with a distinct **branch color**.
4. Each clue attached as a node under **Clues**, with the full clue text placed in a **Note** (right-click → Note) rather than in the node label itself.
5. **Icons/flags** on each clue node to mark its status: e.g. ✓ confirmed / supports a suspect, ? unclear or ambiguous, × ruled out.
6. At least **six cross-links** connecting clue nodes to the suspect node(s) they relate to (a clue may link to more than one suspect).
7. A final node titled **"Verdict"** stating who you believe is the thief, connected by a cross-link to the clues that justify it.

2.6 Timing Breakdown for Part 2

Time	Activity
5 min	Scenario briefing read aloud / displayed; questions answered
10 min	Instructor live-demo: build the root node, Suspects and Clues branches, assign one branch color, add one icon, add one Note, and create one cross-link – modeling the exact techniques students already discovered in Part 1
20 min	Independent/pair work: students build their full case map
5 min	2–3 volunteers present their **Verdict** node and justify it using their cross-linked map